UC SAN DIEGO / ECE 251C

4

Sampling and oversampling

Executive summary

The sampling theorem gives exact recovery of a bandlimited signal from noiseless samples. With independent, zero-mean sample noise of variance σ^2, reconstruction restricted to the known signal band has mean-squared error σ^2/r, where $r = \omega_0/(2B)$ is the oversampling factor. We derive this formula by expressing the error as filtered noise, then evaluating the energy of the reconstruction weights using Poisson summation and Parseval's identity. The choice of filter matters: retaining the full Nyquist band also retains noise outside the signal band and gives no such reduction.

4.1 Bandlimited signals and the sampling theorem

We begin by stating precisely the reconstruction result obtained in Lecture 3. The bandwidth assumption identifies the class in which the samples determine a unique signal; it will also tell us which frequencies to retain when the samples are noisy.

Definition. A signal $f \in L^2(\mathbb{R})$ is *bandlimited to* $B > 0$ if

$$\operatorname{supp} \widehat{f} \subseteq [-B, B], \qquad \text{that is,} \qquad \widehat{f}(\omega) = 0 \quad \text{for } |\omega| > B.$$

Here B is an *angular* frequency. A finite-energy spectrum supported on $[-B, B]$ is also absolutely integrable, since Cauchy–Schwarz gives $\int_{-B}^{B} |\widehat{f}(\omega)|\, \mathrm{d}\omega \le \sqrt{2B}\, \|\widehat{f}\|_{L^2}$. Its inverse Fourier integral therefore defines a continuous function. We use this continuous representative when evaluating the samples $f(nT)$.

Theorem 4.1 (Sampling theorem)**.** Let $f \in L^2(\mathbb{R})$ be bandlimited to B. If $0 < T < \pi/B$, then the samples $f[n] := f(nT)$ determine f uniquely, and

$$\boxed{f(t) = \sum_{n \in \mathbb{Z}} f[n] \operatorname{sinc}\!\left(\frac{t - nT}{T}\right),} \tag{4.1}$$

where $\operatorname{sinc}(s) = \sin(\pi s)/(\pi s)$ for $s \neq 0$ and $\operatorname{sinc}(0) = 1$.

This is the Whittaker–Nyquist–Kotelnikov–Shannon sampling theorem. We recall the spectral argument before introducing noise, since the choice of reconstruction passband will determine the noise reduction.

Conversion to ordinary frequency. For a time signal, write $\omega = 2\pi\nu$, with ν measured in hertz. Set $\widetilde{B} = B/(2\pi)$. Then

$$T < \frac{\pi}{B} \iff T < \frac{1}{2\widetilde{B}} \iff \underbrace{\frac{1}{T}}_{\text{samples per second}} > 2 \underbrace{\widetilde{B}}_{\text{highest frequency in Hz}}. \quad (4.2)$$

Thus the familiar sampling-rate condition in hertz is the same condition written in different units. Throughout the analysis below, B remains the angular-frequency bandlimit.

4.2 Recall: sampling repeats the spectrum

Our Fourier convention is

$$\widehat{f}(\omega) = \int_{\mathbb{R}} f(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t, \qquad f(t) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{f}(\omega)\mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega.$$

The sampled impulse train is

$$f_T(t) = f(t)\text{Ш}_T(t) = \sum_{n\in\mathbb{Z}} f[n]\delta(t-nT), \qquad \omega_0 = \frac{2\pi}{T}.$$

Recall (Poisson summation). The identity from Lecture 3 is

$$\boxed{\widehat{f}_T(\omega) = \sum_{n\in\mathbb{Z}} f[n]\mathrm{e}^{-\mathrm{j}\omega nT} = \frac{1}{T}\sum_{k\in\mathbb{Z}} \widehat{f}(\omega - k\omega_0).} \quad (4.3)$$

The copies of $\widehat{f}$ are separated by ω_0. When $B < \omega_0/2$, only the central copy lies in $[-\pi/T, \pi/T]$.

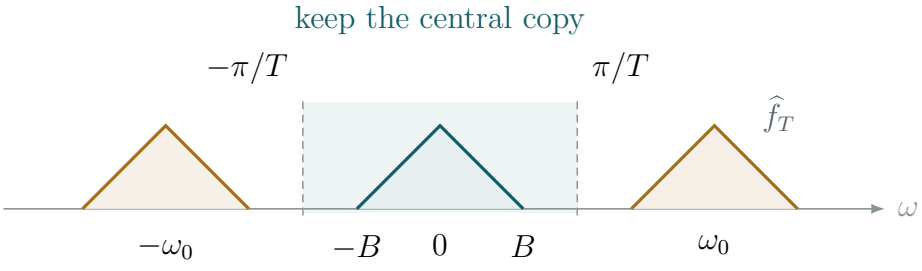


Figure 4.1: The shaded interval contains the original spectral copy and no others. Its amplitude is $\widehat{f}/T$, so reconstruction must also restore the factor T.

Proof of the reconstruction formula. Choose

$$\widehat{h}_T(\omega) = T\mathbf{1}_{[-\pi/T,\pi/T]}(\omega).$$

Then $\widehat{h}_T\widehat{f}_T = \widehat{f}$. The inverse transform of this rectangle is, for $t \neq 0$,

$$\begin{aligned} h_T(t) &= \frac{T}{2\pi}\int_{-\pi/T}^{\pi/T} \mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega \\ &= \frac{T}{2\pi}\frac{\mathrm{e}^{\mathrm{j}\pi t/T} - \mathrm{e}^{-\mathrm{j}\pi t/T}}{\mathrm{j}t} = \operatorname{sinc}(t/T). \end{aligned}$$

At $t = 0$, the integral equals one. Convolution with f_T gives

$$(h_T * f_T)(t) = \sum_{n\in\mathbb{Z}} f[n]h_T(t - nT) = f(t).$$

This proves the reconstruction identity in equation (4.1). It also proves uniqueness within the specified bandlimited class: two signals with the same samples have the same reconstructed spectrum and hence the same continuous representative.□

4.3 Oversampling and noisy measurements

Exact recovery does not distinguish between different sampling steps that satisfy the bandwidth condition. To study their effect on noise, we keep the signal class fixed while changing the sample spacing.

Setup. Keep the bandlimit B fixed and choose a sampling step satisfying

$$\omega_0 = \frac{2\pi}{T} \geq 2B, \qquad r := \frac{\omega_0}{2B} = \frac{\pi}{BT} \geq 1. \tag{4.4}$$

The ratio r is the *oversampling factor*. It compares the sampling angular frequency with the width $2B$ of the signal band. At $r = 1$, neighboring spectral copies meet only at their endpoints; at $r > 1$, there is a gap between them. The endpoint contact is harmless for finite-energy spectra, because changing values at finitely many frequencies does not change a Fourier integral.

Measurements. Instead of the exact samples, we observe

$$y[n] = f[n] + w[n], \qquad \mathbb{E}\{w[n]\} = 0, \qquad \mathbb{E}\{|w[n]|^2\} = \sigma^2. \tag{4.5}$$

Assume the $w[n]$ are independent and identically distributed. If $n \neq m$, independence and zero mean give

$$\mathbb{E}\{w[n]w^*[m]\} = \mathbb{E}\{w[n]\}\,\mathbb{E}\{w^*[m]\} = 0.$$

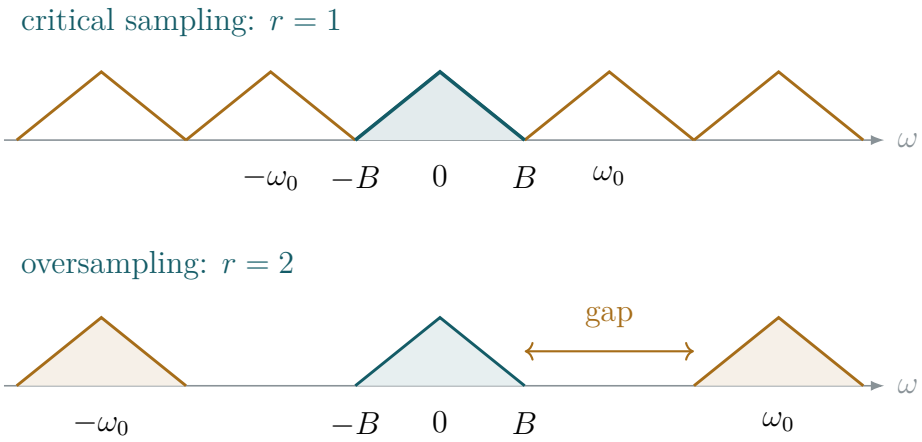


Figure 4.2: The signal band stays fixed while the spectral copies move apart. Both rows show $T\widehat{f}_T$, so their heights are comparable even though the sampling steps differ.

If $n = m$, the expectation is σ^2. In one formula,

$$\mathbb{E}\{w[n]w^*[m]\} = \sigma^2\delta[n-m]. \tag{4.6}$$

The calculation below uses only the zero mean and the covariance identity (4.6); a Gaussian distribution is not required. When comparing different sampling rates, we keep the variance σ^2 of each sample's noise fixed. This is an assumption of the model, not a general property of every acquisition device.

4.4 Reconstruct only the signal band

The cardinal reconstruction filter retains every frequency in $[-\pi/T, \pi/T]$. Under oversampling, part of that interval lies outside the known signal band. Noiseless reconstruction does not require those additional frequencies. We can therefore choose the narrower *signal-band* response

$$\boxed{\widehat{h}_{\mathrm{LP}}(\omega) = T\mathbf{1}_{[-B,B]}(\omega).} \tag{4.7}$$

It still satisfies $\widehat{h}_{\mathrm{LP}}\widehat{f}_T = \widehat{f}$: it keeps the entire central copy, removes the others, and cancels the sampling factor $1/T$.

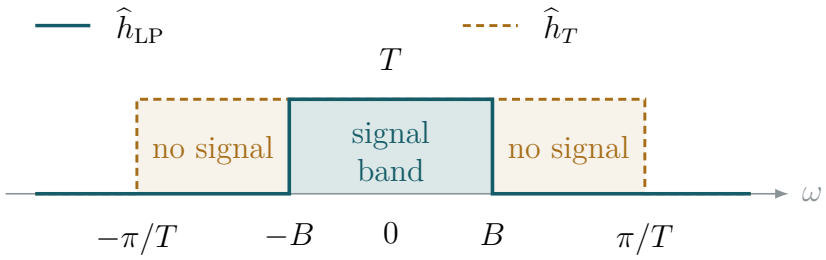


Figure 4.3: Two filters reconstruct the same noiseless signal. The solid response rejects the extra frequency bands admitted by the dashed, full-Nyquist response. With noisy samples, that distinction matters.

Calculate the impulse response. Inverting equation (4.7), for $t \neq 0$,

$$\begin{aligned} h_{\mathrm{LP}}(t) &= \frac{1}{2\pi} \int_{-B}^{B} T \mathrm{e}^{\mathrm{j}\omega t} \, \mathrm{d}\omega \\ &= \frac{T}{2\pi} \left[\frac{\mathrm{e}^{\mathrm{j}\omega t}}{\mathrm{j}t} \right]_{-B}^{B} = \frac{T}{2\pi} \frac{2\mathrm{j}\sin(Bt)}{\mathrm{j}t} \\ &= \frac{T \sin(Bt)}{\pi t} = \frac{BT}{\pi} \operatorname{sinc}\left(\frac{Bt}{\pi} \right). \end{aligned} \tag{4.8}$$

At zero, the integral gives $h_{\mathrm{LP}}(0) = BT/\pi = 1/r$. Thus

$$h_{\mathrm{LP}}(t) = \frac{1}{r} \operatorname{sinc}\left(\frac{t}{rT} \right).$$

At $r = 1$, this is the cardinal kernel $h_T(t) = \operatorname{sinc}(t/T)$. At $r > 1$, the signal-band kernel is different and generally does not vanish at the other sample locations. This does not contradict exact reconstruction: the oversampled values are constrained by the bandwidth assumption, and their combined contributions reproduce f. The individual reconstruction functions need not themselves be cardinal interpolants.

4.5 The reconstruction error is filtered noise

Apply the signal-band reconstruction kernel to the noisy measurements, and denote the resulting estimate by

$$\bar{f}(t) \coloneqq \sum_{n \in \mathbb{Z}} y[n] h_{\mathrm{LP}}(t - nT). \tag{4.9}$$

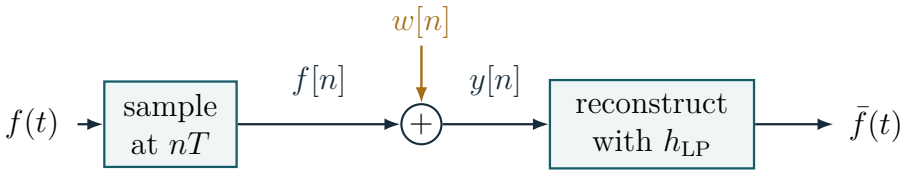


Figure 4.4: Noise is added to the samples. The reconstruction filter acts on both the signal samples and the noise samples.

Substitute $y[n] = f[n] + w[n]$ and use exact noiseless reconstruction:

$$\begin{aligned}\bar{f}(t) &= \sum_n \big(f[n] + w[n]\big) h_{\mathrm{LP}}(t - nT) \\ &= \underbrace{\sum_n f[n] h_{\mathrm{LP}}(t - nT)}_{f(t)} + \sum_n w[n] h_{\mathrm{LP}}(t - nT).\end{aligned}$$

The bandwidth assumption identifies the first sum with $f(t)$. The second sum is the response of the reconstruction kernel to the sample noise, so it accounts for all of the reconstruction error. In particular,

$$e(t) \coloneqq \bar{f}(t) - f(t) = \sum_{n \in \mathbb{Z}} w[n] h_{\mathrm{LP}}(t - nT). \tag{4.10}$$

The noise series is interpreted in mean square at each fixed t. The sum of squared weights will be shown to be finite below, which guarantees this convergence. We may therefore perform the next calculation first for finite sums and then pass to their mean-square limit; no finite energy of the entire noise sequence is assumed.

Mean-squared error. Using $|z|^2 = zz^*$ and linearity of expectation gives

$$\begin{aligned}\mathbb{E}\{|e(t)|^2\} &= \mathbb{E}\left\{ \left| \sum_{n \in \mathbb{Z}} w[n] h_{\mathrm{LP}}(t - nT) \right|^2 \right\} \\ &= \sum_{n,m \in \mathbb{Z}} \mathbb{E}\{w[n] w^*[m]\}\, h_{\mathrm{LP}}(t - nT) h_{\mathrm{LP}}^*(t - mT).\end{aligned}$$

By equation (4.6), the terms with $n \neq m$ vanish. Each surviving term has $n = m$ and contributes $\sigma^2 |h_{\mathrm{LP}}(t - nT)|^2$. Therefore

$$\mathbb{E}\{|e(t)|^2\} = \sigma^2 \sum_{n \in \mathbb{Z}} |h_{\mathrm{LP}}(t - nT)|^2. \tag{4.11}$$

Thus the probabilistic part of the problem has reduced to a deterministic one: compute the squared ℓ^2 norm of the reconstruction weights. Poisson summation identifies their spectrum, and Parseval's identity then evaluates the sum.

4.6 A reflected and shifted reconstruction kernel

Fix the location t at which we want to evaluate the error. Define

$$a_t[n] \coloneqq h_{\mathrm{LP}}(t - nT), \qquad g_t(u) \coloneqq h_{\mathrm{LP}}(t - u). \tag{4.12}$$

The sequence a_t contains the weights used to reconstruct at the fixed location t. It is obtained by sampling g_t, since $a_t[n] = g_t(nT)$. In the definition of g_t, the variable is u; the subscript t records which reconstruction location has been fixed. Keeping these roles separate is essential for the change of variables below.

Calculate the Fourier transform of g_t. Set

$$v = t - u, \qquad u = t - v, \qquad \mathrm{d}u = -\mathrm{d}v.$$

The integration limits change as well:

$$u \to -\infty \implies v \to +\infty, \qquad u \to +\infty \implies v \to -\infty.$$

Writing both changes explicitly gives

$$\begin{aligned}
\widehat{g}_t(\omega) &= \int_{-\infty}^{\infty} h_{\mathrm{LP}}(t-u)\mathrm{e}^{-\mathrm{j}\omega u}\,\mathrm{d}u \\
&= -\int_{+\infty}^{-\infty} h_{\mathrm{LP}}(v)\mathrm{e}^{-\mathrm{j}\omega(t-v)}\,\mathrm{d}v \\
&= \int_{-\infty}^{\infty} h_{\mathrm{LP}}(v)\mathrm{e}^{-\mathrm{j}\omega(t-v)}\,\mathrm{d}v \\
&= \mathrm{e}^{-\mathrm{j}\omega t}\int_{-\infty}^{\infty} h_{\mathrm{LP}}(v)\mathrm{e}^{\mathrm{j}\omega v}\,\mathrm{d}v \\
&= \mathrm{e}^{-\mathrm{j}\omega t}\widehat{h}_{\mathrm{LP}}(-\omega).
\end{aligned} \tag{4.13}$$

The minus sign from $\mathrm{d}u = -\mathrm{d}v$ cancels the minus sign from reversing the limits. The positive exponential inside the last integral changes the *frequency argument* to $-\omega$; it does not introduce a minus sign in front of the transform.

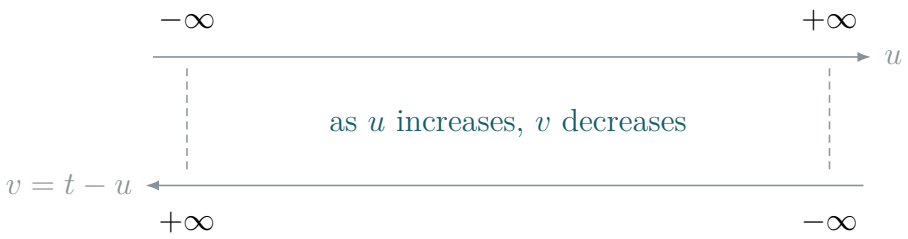


Figure 4.5: The substitution reverses orientation. Keeping the original integration limits while inserting only $\mathrm{d}u = -\mathrm{d}v$ would create an incorrect extra minus sign.

4.7 Apply Poisson summation to the weights

The auxiliary function g_t has converted the reconstruction weights into samples of a bandlimited function. We can therefore apply the same sampling identity used for f, now to compute the spectrum of the weights. Use physical angular frequency to write the DTFT of a_t:

$$A_t(\omega) \coloneqq \sum_{n\in\mathbb{Z}} a_t[n]\mathrm{e}^{-\mathrm{j}\omega nT}. \tag{4.14}$$

It is ω_0-periodic, since $\mathrm{e}^{-\mathrm{j}\omega_0 nT} = \mathrm{e}^{-\mathrm{j}2\pi n} = 1$.

Recall (Poisson summation). For a bandlimited finite-energy function g with samples $g[n] = g(nT)$,

$$\sum_{n\in\mathbb{Z}} g[n]\mathrm{e}^{-\mathrm{j}\omega nT} = \frac{1}{T}\sum_{k\in\mathbb{Z}} \widehat{g}(\omega - k\omega_0), \qquad \omega_0 = \frac{2\pi}{T}. \tag{4.15}$$

Apply this identity to the bandlimited function $g = g_t$ from equation (4.12). Substituting equation (4.13) gives

$$\begin{aligned} A_t(\omega) &= \frac{1}{T}\sum_{k\in\mathbb{Z}} \widehat{g}_t(\omega - k\omega_0) \\ &= \frac{1}{T}\sum_{k\in\mathbb{Z}} \widehat{h}_{\mathrm{LP}}(-\omega + k\omega_0)\mathrm{e}^{-\mathrm{j}(\omega - k\omega_0)t}. \end{aligned} \tag{4.16}$$

Restricting to one period. If $|\omega| < \omega_0/2$ and $k \neq 0$, then

$$|-\omega + k\omega_0| \geq |k|\omega_0 - |\omega| > \frac{\omega_0}{2} \geq B.$$

The corresponding filter value is zero. Only $k = 0$ remains, so on the base interval

$$\boxed{A_t(\omega) = \frac{1}{T}\widehat{h}_{\mathrm{LP}}(-\omega)\mathrm{e}^{-\mathrm{j}\omega t} = \mathbf{1}_{[-B,B]}(\omega)\mathrm{e}^{-\mathrm{j}\omega t}.} \tag{4.17}$$

Here $\widehat{h}_{\mathrm{LP}}(-\omega) = T\mathbf{1}_{[-B,B]}(\omega)$ because the passband is symmetric. Endpoints do not affect the energy calculation.

4.8 Parseval: track the frequency normalization

The ordinary DTFT uses a dimensionless angular frequency θ:

$$\sum_{n\in\mathbb{Z}} a_t[n]\mathrm{e}^{-\mathrm{j}\theta n} = A_t(\theta/T), \qquad \theta = \omega T.$$

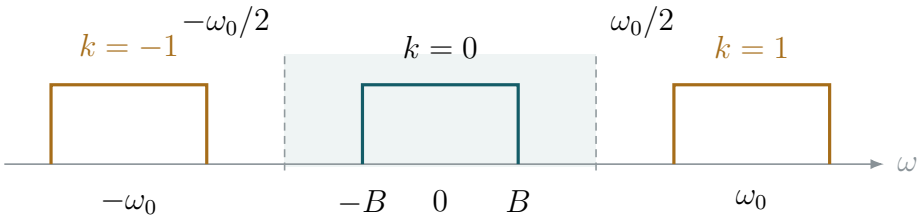


Figure 4.6: Magnitudes of three terms in the periodization. The shaded base interval intersects only the central rectangle. The phase factor depends on t, but its magnitude is always one.

Recall (Parseval). Sequence energy equals the normalized integral of squared DTFT magnitude over one period:

$$\sum_{n\in\mathbb{Z}} |a_t[n]|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |A_t(\theta/T)|^2 \,\mathrm{d}\theta. \tag{4.18}$$

Substitute $\theta = \omega T$, with $\mathrm{d}\theta = T\,\mathrm{d}\omega$. Since $\omega_0 = 2\pi/T$, the new integration limits are $-\omega_0/2$ and $\omega_0/2$. Hence

$$\begin{aligned}\sum_{n} |a_t[n]|^2 &= \frac{1}{2\pi}\int_{-\pi}^{\pi} |A_t(\theta/T)|^2 \,\mathrm{d}\theta \\ &= \frac{T}{2\pi}\int_{-\omega_0/2}^{\omega_0/2} |A_t(\omega)|^2 \,\mathrm{d}\omega \\ &= \frac{T}{2\pi}\int_{-B}^{B} 1 \,\mathrm{d}\omega = \frac{BT}{\pi} = \frac{1}{r}.\end{aligned} \tag{4.19}$$

The third line uses equation (4.17). The dependence on t is a phase factor of magnitude one, so it disappears from the energy calculation. Consequently, the squared norm of the weights is the same at every reconstruction location. In particular, the sum is finite, as required for the mean-square interpretation of the noise series.

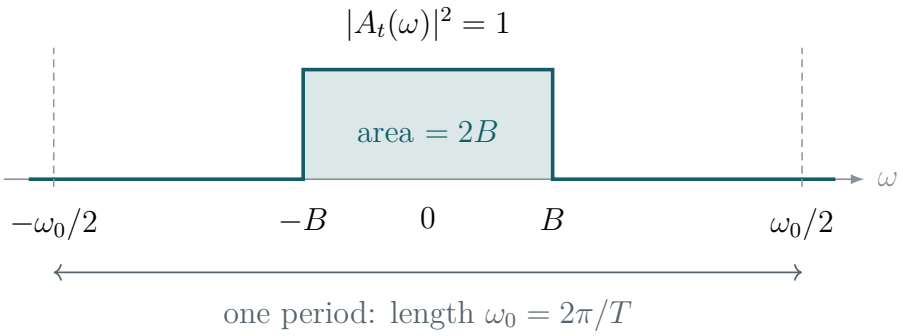


Figure 4.7: Parseval averages this squared magnitude over a period. The occupied fraction is $2B/\omega_0 = BT/\pi = 1/r$. Increasing the sampling rate enlarges the period, not the signal band.

4.9 The reconstruction error falls as $1/r$

Combining equations (4.11) and (4.19) gives the exact error variance:

$$\boxed{\mathbb{E}\{|\bar{f}(t) - f(t)|^2\} = \sigma^2 \frac{BT}{\pi} = \frac{\sigma^2}{r}, \qquad t \in \mathbb{R}.} \tag{4.20}$$

The error has mean zero because the sample noise has mean zero. Its mean-squared size is therefore also its variance. The expectation averages over noise realizations at a *fixed location*; it is not an integral of the error over the whole line.

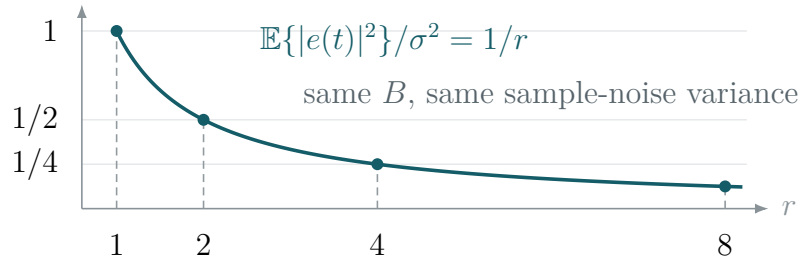


Figure 4.8: The exact normalized mean-squared error, not an empirical fit. Doubling the oversampling factor halves the variance. The root-mean-squared error decreases by a factor of $\sqrt{2}$.

Example. Suppose the per-sample noise standard deviation is $\sigma = 0.1$. Then $\sigma^2 = 0.01$. At $r = 2$, the reconstructed error variance is 0.005; at $r = 4$, it is 0.0025, with standard deviation 0.05.

Why the filter matters. As a worked consequence, suppose instead that we reconstruct using the full-Nyquist kernel $h_T(t) = \mathrm{sinc}(t/T)$. Repeating the same calculation with cutoff π/T gives

$$\sum_n |h_T(t - nT)|^2 = \frac{T}{2\pi} \int_{-\pi/T}^{\pi/T} 1 \, \mathrm{d}\omega = 1.$$

The resulting error variance is σ^2, not σ^2/r. Both kernels reconstruct the noiseless signal exactly, but the full-Nyquist kernel also retains frequencies that the signal model says contain no signal. The noise reduction is therefore due to oversampling *together with* reconstruction restricted to the known signal band.

> **What oversampling buys**
>
> With the signal band and per-sample noise variance fixed, more samples allow the signal-band reconstruction filter to reject a larger fraction of the noise. The improvement is a mean-squared-error statement under the specified independent-noise model, not a guarantee for every individual noise realization.